



Student CyberGuard Guidebook

Quick-Reference Guide for Digital Classrooms, MFA, and Secrets Management

Welcome to your digital classroom. As education increasingly shifts to cloud platforms, virtual labs, and collaborative digital environments, cybersecurity has become a fundamental student responsibility. This guide provides actionable steps to secure your academic identity, protect sensitive project credentials, and safely navigate potential scams.

1. Configuring Multi-Factor Authentication (MFA)

Multi-Factor Authentication (MFA) adds a critical layer of defense beyond your password. Even if a malicious actor acquires your login credentials, they cannot access your account without a secondary, time-sensitive verification token.

Standard Setup Process:

- 1. Access Security Settings:** Log into your student portal, navigate to your account profile or settings, and look for "Security Settings" or "Two-Step Verification".
- 2. Choose an Authenticator Method:** Select *Authenticator App* as your primary method. Applications like Google Authenticator or Microsoft Authenticator are preferred over SMS verification.
- 3. Link the Account:** Scan the displayed QR code using your smartphone's authenticator app to initialize secure, time-based one-time passwords (TOTP).
- 4. Secure Your Backup Codes:** Immediately save the provided single-use recovery/backup codes. Store them in a secure offline location or an encrypted vault. If you lose access to your device, these codes are the only way to recover your account without institutional IT intervention.

Pro Tip: Beware of MFA Fatigue

If you receive unexpected push notifications to authorize a login that you did not initiate, **deny the request immediately**. Attackers often flood a target's phone with authorization prompts hoping the user will accidentally tap accept out of frustration or distraction.

2. Managing Secrets, Passwords & Developer Tokens

As a student, you handle diverse sets of credentials, including Learning Management System (LMS) access, remote laboratory credentials, and potentially cloud provider API keys for programming assignments. Proper management prevents catastrophic leaks and unauthorized resource utilization.

Passphrases vs. Passwords

Avoid traditional passwords containing complex substitutions like `P@$$w0rd!` which are easily predicted by modern brute-force algorithms. Instead, construct **passphrases**: long, distinct combinations of four or more random words (e.g., `correct-horse-battery-staple`). These are easy for humans to memorize but extremely computationally expensive to crack.

Preventing Developer Credential Leaks

If your coursework involves software engineering or data analysis, handling production secrets requires strict protocols:

- **Never Hardcode Secrets:** Do not insert API tokens, database keys, or SSH passwords directly into your source code.
- **Utilize Environment Variables:** Store credentials locally in external system environment variables or inside a dedicated configuration file like `.env`.
- **Configure Exclusion Rules:** Always add your secrets files to a `.gitignore` file before staging files. Committing an API token to public platforms like GitHub can result in malicious automated scripts draining cloud credits within minutes.

3. Detecting and Avoiding Classroom Scams

Digital classrooms are high-value environments for social engineers and scammers targetting student financial aid, data, and institutional access.

Common Attack Vectors

- **Phishing via Student Email:** Messages impersonating professors or the bursar's office demanding urgent updates to financial details or immediate tuition payment.
- **Malicious Syllabus Downloads:** Files shared on informal community groups (e.g., Discord, WhatsApp) claiming to be "free textbooks" or "exam answers" containing embedded malware.
- **Fake LMS Landing Pages:** Spoofed login portals mirrored to capture your school login credentials when logging into systems like Canvas or Blackboard.

Defensive Measures

- **Verify URLs Closely:** Always check the top-level domain before providing account details. Ensure it matches your school's exact structure (e.g., `.edu`).
- **Inspect Header Info:** Check the actual email sender address, not just the display name. Real academic communications will rarely originate from external domains like Gmail.
- **Isolate Study Spaces:** Keep personal data separate from educational code environments to minimize potential cross-compromise.

Red Flag Alert: Urgent Direct Messages

Be suspicious of peers or automated bots inside classroom messaging channels asking you to install browser extensions or software utilities to "participate in peer reviews" or "unlock assignment rubrics." This is a primary deployment method for information-stealing malware.

4. Daily Digital Hygiene Checklist

Incorporate these core habits into your daily educational routine to maintain a zero-trust standard:

- ☐ **Verify Active Sessions:** Log out of school portals when using shared campus workstations or public network zones.
- ☐ **Review Code Repositories:** Periodically check repository public/private visibility status before publishing student projects.
- ☐ **Enforce Zero Password Reuse:** Use a dedicated password manager to generate distinct credentials for every academic service.
- ☐ **Report Security Incidents:** Instantly forward any suspicious links or phishing attempts to your institution's central IT helpdesk.